Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ________________
Topic: Applications of Systems of Inequalities **Date:** __________

Foundation (Jalapeno)

Open Ended Questions (FRQ)

Q9. A space station has two solar panels, each with a power output of x watts. The station also has a backup battery with a power output of y watts. The total power output must be at least 1000 watts, and the cost of the solar panels is

$$2x$$

and the cost of the battery is

$$3y$$

. If the budget is

$$6000$$

, find the constraint on x and y. Show your work and explain your answer.

Q10. A spaceship is traveling from Earth to Mars, and its distance d from Earth is given by $d = 2x + 3y$, where x is the time in hours and y is the speed in kilometers per hour. The spaceship must travel at least 4000 kilometers, and the maximum speed is 20 kilometers per hour. Find the minimum time required for the trip, and show your work. Explain your answer in the context of the problem.

Chef's Tips

- Tip 1: Make sure to read each question carefully and understand the context.
- Tip 2: Use the given information to identify the constraints and the objective.
- Tip 3: Show your work and explain your answer for the free-response questions.